DO NOW — RECALL

Lesson 113 · Open Source and Closed Source Decisions

Name: _____ Class: _____ Date: _____

1. Name a utility for recovering files after hardware loss.

2. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

3. What is the time complexity of sequential search in the worst case?

4. After pushing A then B onto a stack, which is popped first?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Backup and restore utility.

2. Model answer

X is removed; Y remains at the front.

3. Model answer

O(n).

4. Model answer

B, because a stack is LIFO.

By the end, you can…

• Open-source licences allow access to source code with specified reuse rights.

• Closed-source vendors control release; support and integration vary by product.

• Evaluate total cost, auditability, maintenance, expertise and obligations.

Vocabulary — know these terms

open source — Code distributed under a licence allowing inspection and specified reuse.

closed source — Source and modification rights normally controlled by owner.

licence — Legal conditions attached to software use.

vendor lock-in — Difficult or expensive dependence on one supplier.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

A proprietary supported package may offer a service contract and managed upgrades; assess data export, recurring licensing, vendor lock-in and contract quality. Source model alone does not decide security.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Source availability

Open-source licences allow access to source code with specified reuse rights.

Control vs support

Closed-source vendors control release; support and integration vary by product.

Make a justified choice

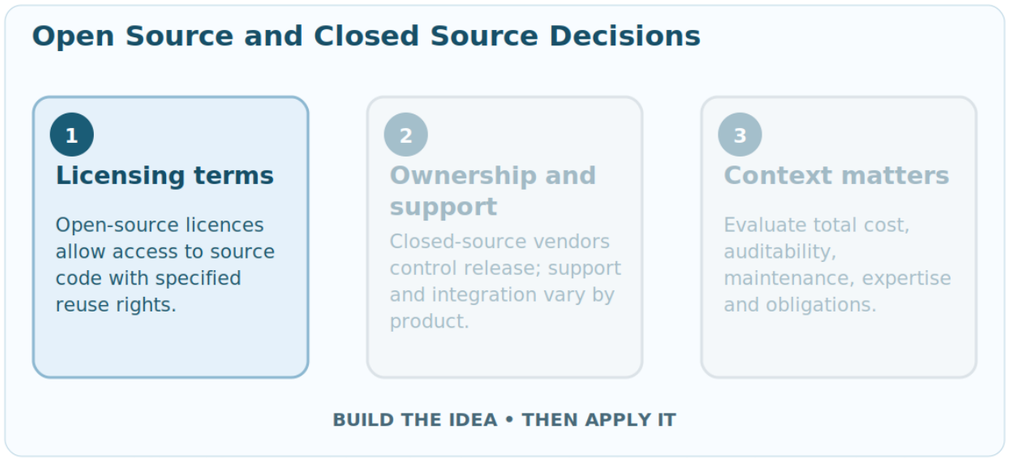
Evaluate total cost, auditability, maintenance, expertise and obligations.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

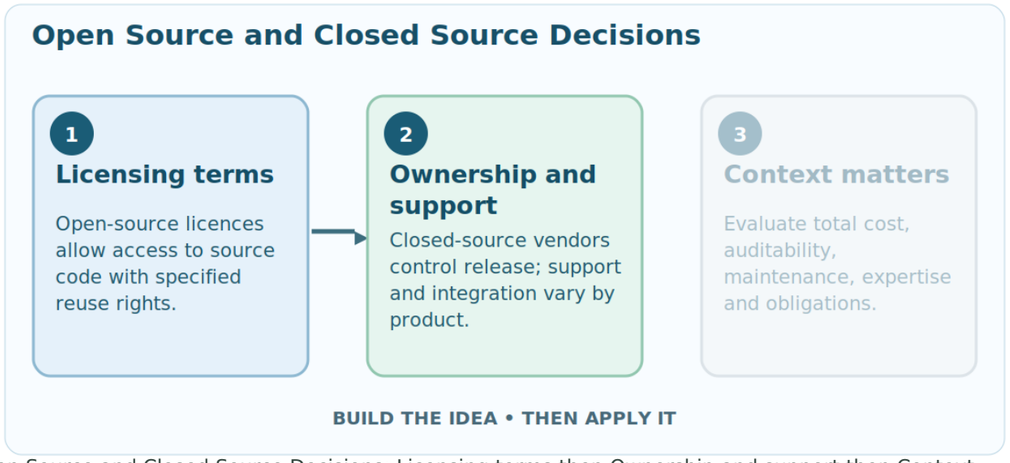
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Open Source and Closed Source Decisions: Licensing terms then Ownership and support then Context matters

New Learning



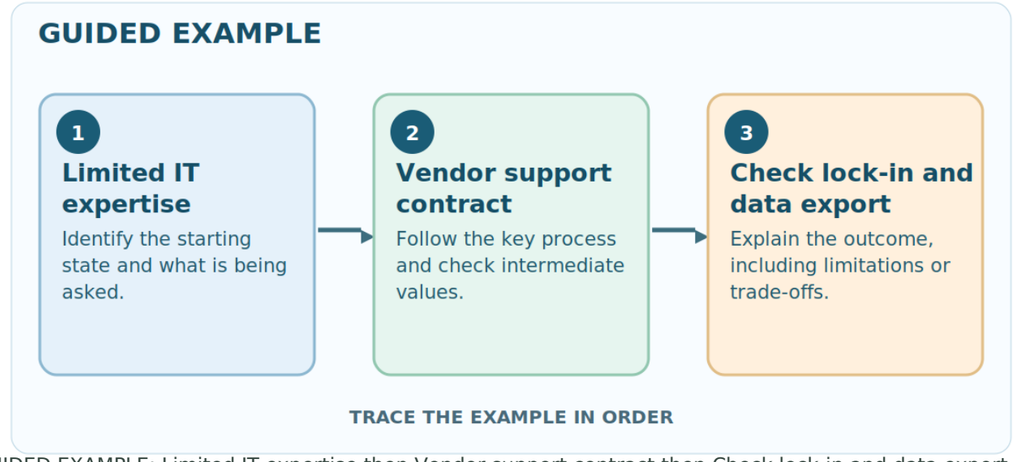
Open Source and Closed Source Decisions: Licensing terms then Ownership and support then Context matters

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A small medical practice needs guaranteed vendor response and has no developers. Compare closed-source options

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Limited IT expertise then Vendor support contract then Check lock-in and data export

We Do Check · original lesson diagrams

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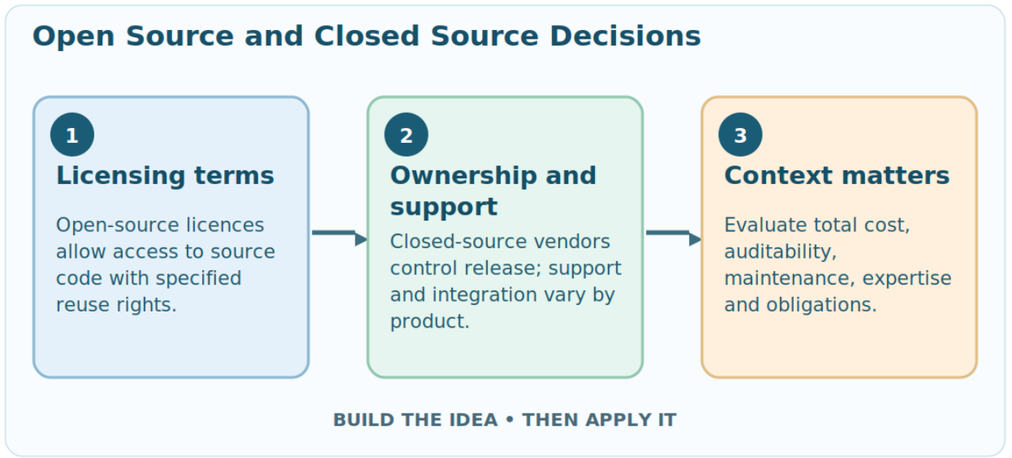


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New Learning



Open Source and Closed Source Decisions: Licensing terms then Ownership and support then Context matters

I DO — WORKED EXAMPLE

Follow the worked method

A council wants to adapt an accessible web service using its development team. Explain an open-source advantage and a risk.

Model answer

Relevant open-source licence can permit modification and inspection, allowing accessibility changes. Risk: the council must ensure support, security patching and licence compliance; source availability alone guarantees neither quality nor safety.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Open-source software is automatically more secure than closed-source software.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

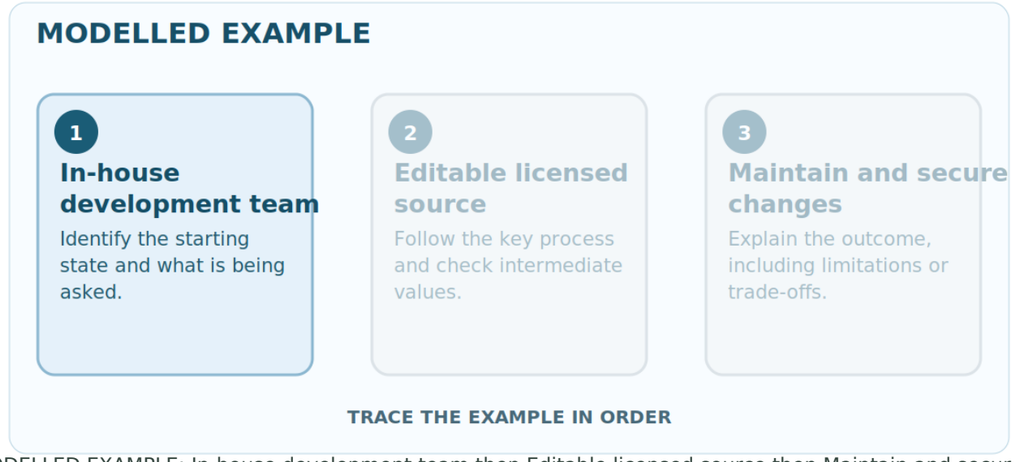
Evaluate the choice for a charity's public website considering skills, privacy, costs and vendor dependence. [8]

I Do · original lesson diagrams

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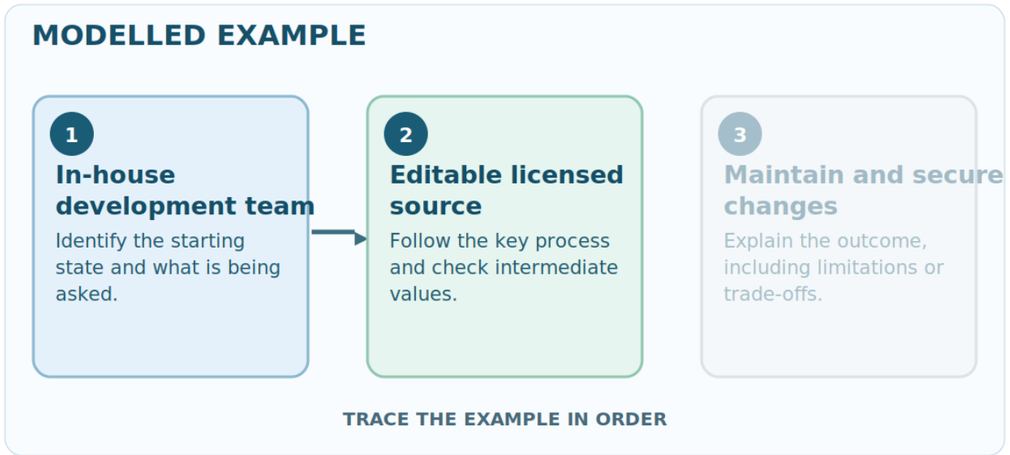
MODELLED EXAMPLE: In-house development team then Editable licensed source then Maintain and secure changes

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YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define open-source and closed-source software and state a licensing implication. [6]

Question 2

Explain why free-of-charge software is not necessarily open source. [4]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

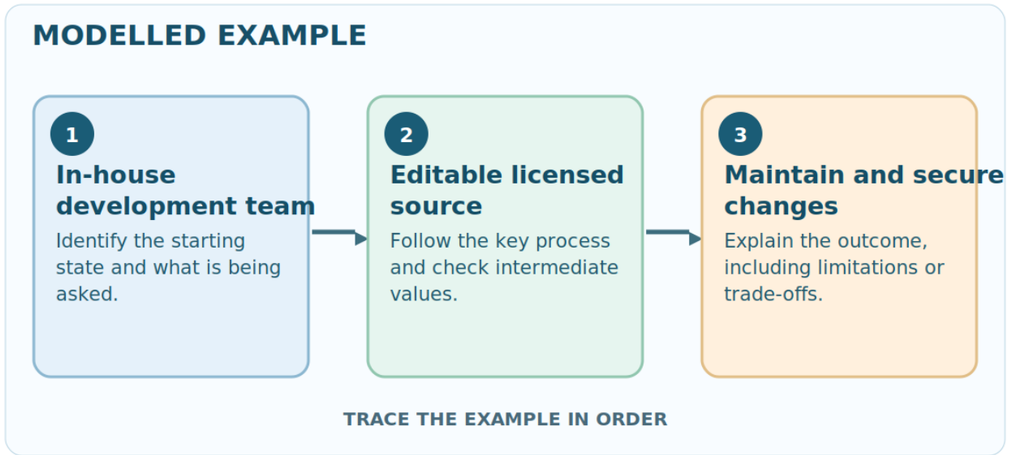
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

Try → Discuss → Improve

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